



On the usefulness of passive seismic imaging for mineral exploration - A case study in the Irish Midlands

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ABSTRACT

In the past years, the demand for critical raw materials, which are needed for green technologies like electric cars, solar panels, and electro-mobility, has rapidly increased. Whilst recycling can provide an increasing portion of these elements, recycling alone cannot meet the estimated demand. This implies that further mining will be required if the EU is to meet its climate goals.

Within the project VECTOR, we are developing and testing a new geophysical technique, which is passive and non-disruptive, to image the Earth's subsurface and structural layers. This method, known as ambient noise tomography, uses signals present in the survey area, such as noise from wind or traffic, to derive a 3D subsurface model. We test the usefulness of this method in Collinstown in the Irish Midlands, known for its significant zinc deposits. Although the resolution of the passive techniques is lower than that of the active seismic methods, we can image layers like the Pale Beds, which are the most prospective host rocks for base metal mineralization, and their depths agree well with those derived from the active 2D methods. This demonstrates that passive seismic surveys can complement active methods, opening new exploration strategies.

1. Introduction

Europe set out its goals for decarbonization in the EU Green Deal, which includes reducing net greenhouse gas emissions by relying more on renewable energy and green technologies. One goal of the EU project VECTOR (<https://vectorproject.eu>) is to test and develop passive, non-disruptive exploration methods to investigate Europe's raw material potential.

The Irish Midlands basin is a globally significant area for zinc exploration and makes Ireland one of the major European producers of zinc, the fourth most consumed metal globally (Watari et al., 2021). Its corrosion resistance properties contribute significantly to the longevity, reliability, and sustainability of construction and infrastructure projects, vehicle manufacturing, and green technologies.

Ireland has a long history of deposit discoveries. After the discovery of the lead–zinc Tynagh deposit in 1962, the increased exploration activity resulted in several other deposits and prospects being discovered throughout the past six decades. The latest major discovery was the Tara Deep Zn–Pb deposit at Navan, County Meath, 50 km northwest of Dublin, which is to date the largest zinc mine in Europe. Navan has produced substantial quantities of zinc and lead ores for

the past 45 years and has played a pivotal role in the economic prosperity of the surrounding area (Redmond, 2023; Ashton et al., 2018). After decades of mineral exploration, the potential for discoveries in the Irish Midlands basin remains high. The main reason is that substantial progress has been made in exploration technologies, including geochemistry, Induced Polarization, electromagnetic surveys, and the deep drilling of geologically defined targets (Ashton et al., 2023). Additionally, exploration has increasingly used 2D and local 3D active seismic surveying, providing high-resolution maps of the key lithostratigraphic units down to several kilometers, and has proven to be effective for directly targeting deep mineral deposits, for example, in Sweden (Markovic et al., 2020) and China (Zheng et al., 2024). Despite the success, the method, in particular 3D seismics, has two major drawbacks: (1) the very high costs of the survey, and (2) the complex logistics of operating such invasive sources in urban environments.

In contrast to active seismic methods, passive seismic surveys use seismic signals present in the survey area (noise from wind, traffic, sea waves, etc.) for the imaging of the subsurface instead of using active sources. Some case studies of passive seismic surveys for exploration

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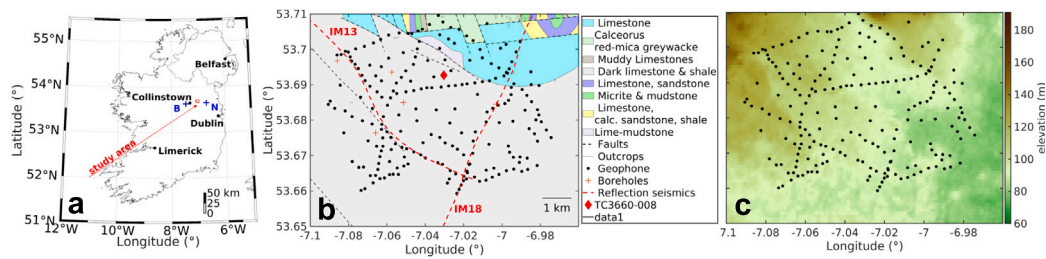


Fig. 1. Station network location in the Irish midlands with approximate positions of deposits at Navan (N) and Ballinalack (B) (left). Geological map (source GSI) with seismic stations in the study area and profile lines (dashed red lines) for the reflection seismic lines IM13 and IM18 provided by Teck Ireland Ltd. ("Teck") (middle). Topography map from NASA's Shuttle Radar Topography Mission SRTM (right) (Farr et al., 2007).

purposes have been carried out, for example, for geothermal exploration (Planès et al., 2019; Martins et al., 2020; Cabrera-Pérez et al., 2023) and mineral exploration (Ryberg et al., 2021; Liu et al., 2022; Li et al., 2023).

In this paper, we describe a passive seismic survey conducted in the Collinstown region in the Irish Midlands to study its usefulness for application in an exploration context. Our study area is located ~27 km west of Navan, between the Navan and the Ballinalack Zn-Pb deposit (Fig. 1). At Navan, over 97% of the ore is hosted by a Courcayan, shallow-water succession termed the Pale Beds (Anderson et al., 1998; Redmond, 2023). The Pale Beds are overlain by the Shaley Pales (shales and carbonates). Overlying the Shaley Pales are the argillaceous bioclastic limestone (ABL) and Waulsortian limestone (Ashton et al., 2018). The mineralization in the Ballinalack deposit, although currently sub-economic, occurs in beds that are stratigraphically equivalent to those that host the Navan deposit (Jones and Brand, 1986). A while ago, a study was undertaken to further the knowledge of the stratigraphy in the Collinstown area (northwest of Navan) and to give a better understanding of the Pale Bed formations there (Nolan, 2014; Philcox, 2018; Sevastopulo, 2015). This study concluded that the Pale Beds formation can be correlated across the Collinstown area, and consequently, it seems likely that there is further base metal mineralization in that area.

We want to test passive seismic methods on their performance in imaging structures in the subsurface, which may play an important role in finding new mineral deposits. The study is carried out within the framework of the EU-funded VECTOR project and also benefits from a large pool of legacy data, which can also be used to help interpret the inversion results.

2. Data and methods

2.1. Seismic network

In November 2022, a network of 215 continuously running digital seismic stations was deployed in an area of $\sim 8 \times \sim 6$ km in the Irish midlands (north of Collinstown), mostly along streets and country roads (Fig. 1). The network consisted of 206 vertical component geophones and 9 broadband sensors. The stations recorded data for ~6 weeks and were recovered in January 2023 (Roy et al., 2024). We used CUBE1 and CUBE3 data recorders with SENSOR Nederland 4.5 Hz vertical geophones and Nanometrics Trillium Compact broadband sensors, respectively. Built-in GPS receivers provided accurate timing.

2.2. Ambient noise tomography

2.2.1. Data selection and processing

The use of ambient noise to extract surface wave empirical Green's functions (EGFs) and to infer Rayleigh group velocities is now well established. Hence, we briefly summarize the procedure here: the data is split into daily segments, which are then one-bit normalized and

spectrally whitened (Bensen et al., 2007). The segments are then cross-correlated and stacked over the recording period of 40 days. This resulted in ~42517 crosscorrelations in total. Fig. 2 shows a virtual shot gather for station 14, i.e., the crosscorrelations of all stations with this station, filtered with three different bandpass filters to evaluate the data quality and the useful noise frequency range. A Rayleigh wave traveling with 2.4 km/s is visible in all shot gathers.

We analyzed the characteristics of the ambient noise, i.e., the frequency content and its directional properties. We checked the directional properties of the noise by plane-wave beamforming in the time domain (Schweitzer et al., 2012; Harjes and Henger, 1973). For the analysis, we used a subset of 18 stations and calculated the beam power of 10-min time windows after bandpass filtering (1–4.5 Hz). Six consecutive beam-forming results were stacked to yield hourly beam-forming plots. Fig. 3 shows examples of these beam-forming plots (normalized beam power) together with a spectrogram (power spectral density; single station) for two arbitrary days (December 27th and 31st 2022).

While the beamforming analysis indicates that the strongest noise energy arrives from two dominant directions, noise energy is still present from all azimuths (greenish area), indicating a sufficiently diffuse wavefield over the analyzed time period. The crosscorrelations were computed by stacking ambient noise recordings over extended periods (~6 weeks), which helps average out temporal variations and directional biases in the noise sources. This long-term stacking ensures a more isotropic and diffuse wavefield, supporting the reliability of the retrieved Green's functions and the subsequent tomography. We have compared daily crosscorrelations between various stations and found that they were stable over time.

Finally, we picked the group velocity dispersion of each receiver pair using the frequency–time analysis (FTAN) method (Dziewonski et al., 1969; Levshin et al., 1972). We did not use the total of 42517 crosscorrelations to determine dispersion curves, but restricted ourselves to inter-station distances greater than 3 km. This is necessary because the inter-receiver distances need to be larger than twice the wavelength at any frequency due to the far-field surface wave approximation that is necessary for ambient-noise surface wave tomography (Yao et al., 2006; Lin et al., 2009). Fig. 4 displays an example of the FTAN analysis.

We selected 2961 crosscorrelation series with a good azimuthal coverage to pick dispersion curves to reduce the computational costs of the Monte Carlo computations, and the redundancy of the raypaths due to the small inter-station distances (400 m). Fig. 5 displays the raycoverage at different periods.

We picked fundamental Rayleigh wave dispersion curves in a semi-automated way using the software of Yao et al. (2006): in the first step, we manually picked dispersion curves for a smaller data subset. In the second step, we calculated an average (reference) dispersion curve from the picked curves. We then allowed the algorithm to automatically pick group velocities around this curve ± 0.5 km/s in the frequency range of 0.55 to 10 Hz. In the final step, the automatically selected

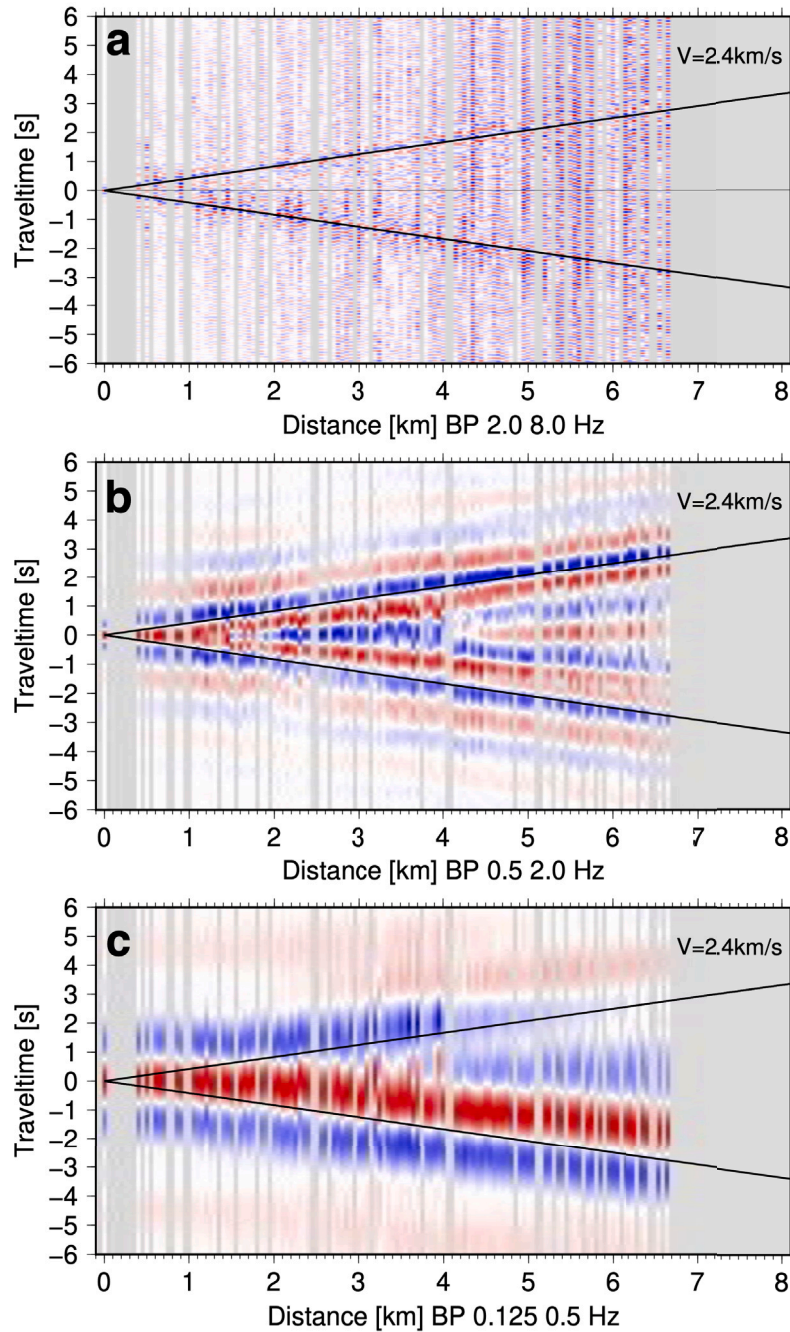


Fig. 2. Virtual shot gather of station 14, filtered with three different bandpass filters. (a) 2–8 Hz, (b) 0.5–2 Hz, and (c) 0.125–0.5 Hz. Traces are sorted by distance and binned. Red and blue colors indicate positive and negative amplitudes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

dispersion curves are cleaned from outliers with unrealistic low group velocities, and we removed all higher-mode arrivals. This resulted in 35,231 data points. Fig. 6 displays all dispersion curves used in the following inversion.

We used group velocity dispersion curves instead of phase velocities, as group velocities are more robustly extracted from ambient noise crosscorrelations. They rely on the arrival of energy envelopes rather than precise phase information, making them less sensitive to noise and processing artifacts. Additionally, phase velocity measurements are affected by 2π phase ambiguity, requiring unwrapping procedures that are sensitive to noise and often unstable, making group velocity a more reliable alternative. We calculate travel times from the group velocities using the interstation distances.

2.2.2. Monte Carlo inversion

We use a probabilistic approach (Zhang et al., 2018, 2020) to invert for 3D subsurface seismic velocity using the picked travel times. The approach is based on a reversible jump Markov chain Monte Carlo algorithm (Green, 1995), which is an iterative stochastic method to generate samples from a target probability density. In a Bayesian approach, all information is described in probabilistic terms. The goal is to calculate the posterior probability distribution function (pdf) which represents the probability of model $\mathbf{m}$ being true given observed data $\mathbf{d}$ and other relevant, priori information. The posterior pdf is defined using Bayes' theorem (Jaynes, 2003): this combines prior knowledge about the model (the prior probability $p(\mathbf{m})$) with a likelihood function $p(\mathbf{d}|\mathbf{m})$ that describes the probability of observing the data if the

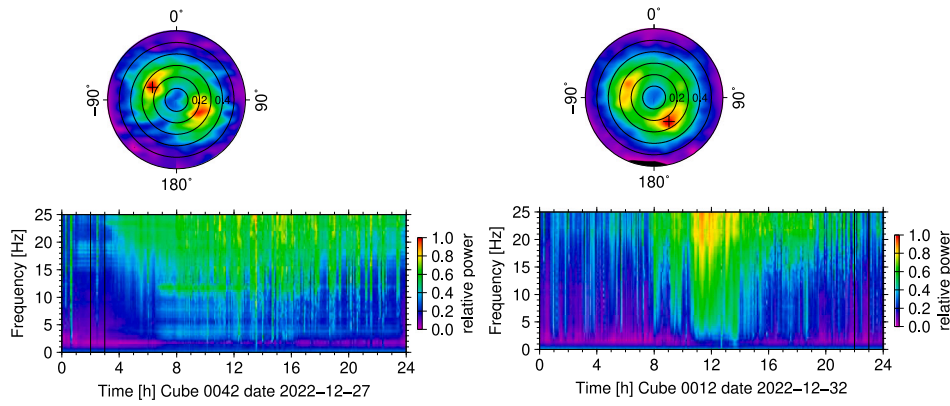


Fig. 3. Frequency and directional beamforming analysis (exemplary). Spectrogram of data at cubes 42 and 12 (bottom) at days 27 and 31 of December 2022 (vertical component). Beamforming analysis in the frequency range 1–4.5 Hz within two time windows of 1 h between 00:00 and 01:00 UTC (left) and 23:00 and 00:00 UTC, respectively (right) using a subset of 18 stations. Normalized beam power is color-coded. Radius is the slowness in skm^{-1} from 0 at the center to 0.6.

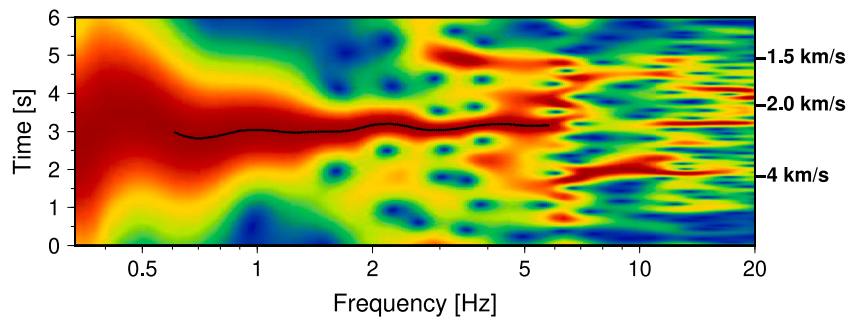


Fig. 4. Example of dispersion curve analysis for a station-pair 054–120 using the FTAN method. The surface wave dispersion curve is picked here between 0.6 and 6 Hz (black solid line). Traces are normalized for every frequency.

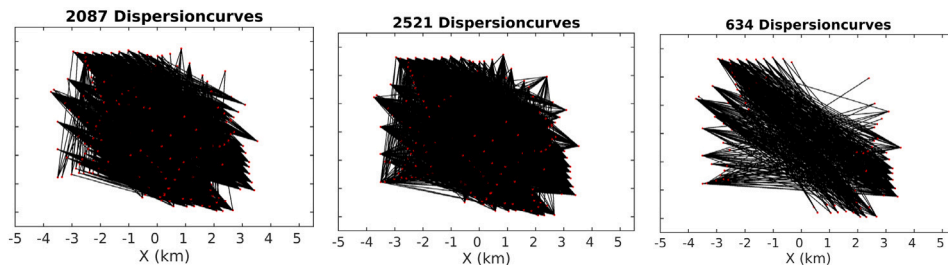


Fig. 5. Raycoverage at 0.1 s (left), 1 s (middle), and 15 s (right).

particular model $\mathbf{m}$ was true). In our approach, the posterior probability is a trans-dimensional function: the number of parameters to describe the 3D model is not fixed, and hence the posterior pdf is defined across many spaces with different dimensionalities.

The 3D subsurface velocity model is defined in terms of a Voronoi tessellation of constant velocity cells, where both the position and velocity values of Voronoi cells and their number can change during sampling, guided by the data and prior information. The P-wave velocity is given by a fixed V_p/V_s ratio with a typical crustal value of 1.73, and the density is calculated following Kurita (1973). We use a two-step forward modeling approach of Zhang et al. (2020) to simulate surface wave dispersion data. In the first step, we extract a local 1D velocity model from the 3D Voronoi model and simulate dispersion curves using the method of Herrmann (2013) for each frequency. Repeating this for all geographical points, we obtain a 2D group velocity map, which can then be used for the calculation of receiver-to-receiver travel times. Here, we modified the original code and replaced the fast marching method Rawlinson and Sambridge (2004) with a faster eikonal solver from Podvin and Lecomte (1991). Computational speed is essential

when using the Markov chain Monte Carlo inversion approach as the forward problem (i.e., calculating the travel times for a given model) has to be solved many ($>10^8$) times.

3. Results

First, we performed checkerboard tests to assess the resolution of our 3D seismic models before inverting the real data. Although they represent an unrealistic pattern of high and low velocities ($\pm 7\%$), they help to understand real data inversion results (Supplementary). The first checkerboard test has tiles of 5×5 km and a grid spacing of 0.25 km in the x and y direction and 0.1 km in depth. The second checkerboard test consists of smaller patches of 1.7×1 km and a grid with $dx = 0.125$ km in horizontal directions and 0.1 km in depth. Figures 1 and 2 in the Supplementary display the results for both synthetic models. The tests show that fast velocity patches (blue patches) are well recovered down to 1.2 km depths in both tests, while the slow velocity perturbations (red patches) are less well resolved and quickly lose resolution with depths. The reason for this is that rays do

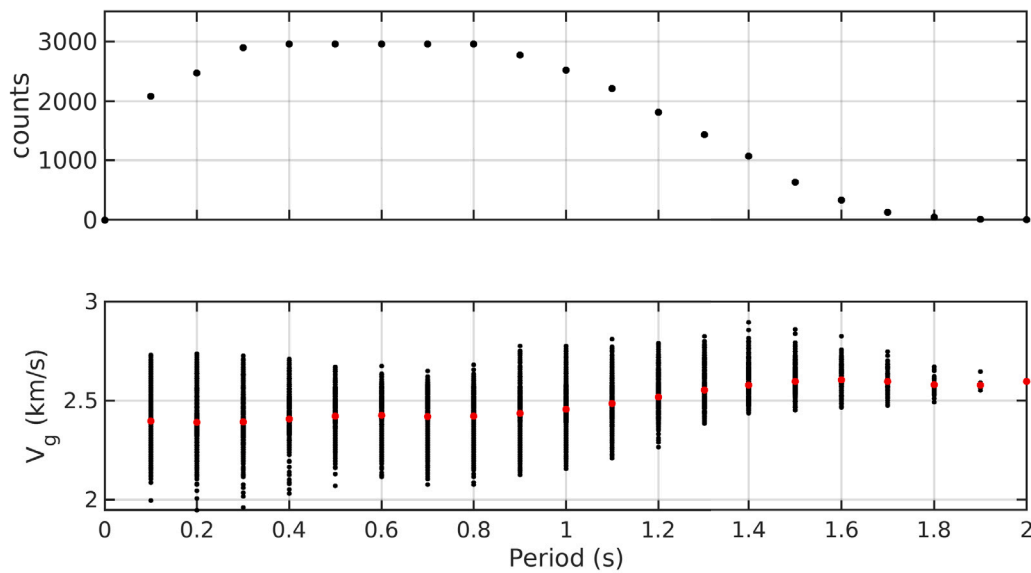


Fig. 6. Data used in the inversion. Number of data points by period (top) and dispersion curves (bottom). Red points mark the mean. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

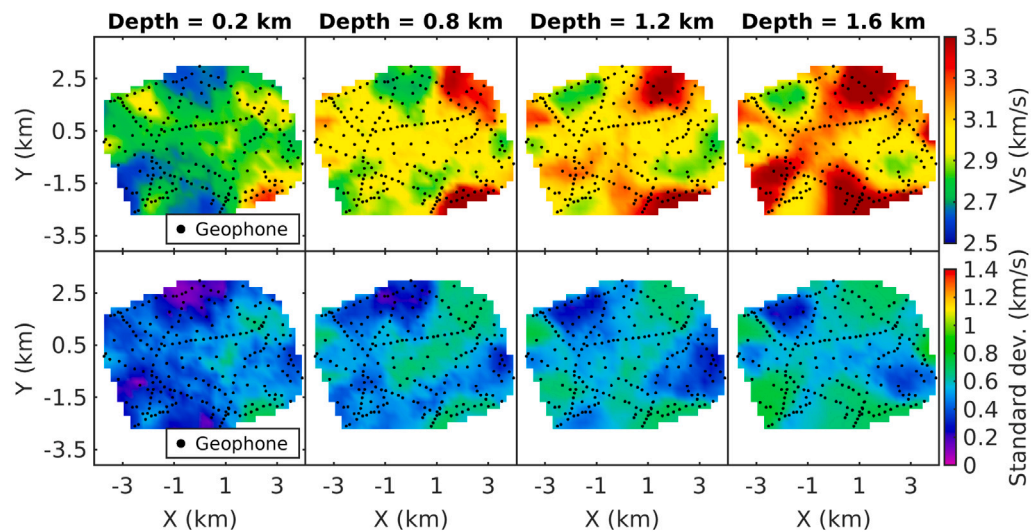


Fig. 7. Depth slices through the average 3D seismic shear wave velocity model (top) and its uncertainties (bottom).

not pass through very low-velocity anomalies but refract around the edges because rays are, by definition, minimum-travel-time paths, well known from controlled source seismics. Hence, the low velocity patches are less well resolved, and the high velocities through which the rays pass are clearly visible. Consequently, the difference between the recovered and true model (bottom row) is smaller for the high velocity perturbations. The edges between the patches have large differences. The smaller perturbations of ~ 1 km in the second checkerboard tests are also resolvable, which demonstrates the potential of our method.

3.1. Ambient noise inversion results for Collinstown

We used the same grid as for the first checkerboard test for the inversion of the real data. The probability distributions to represent the prior constraints on the S velocities are uniform, ranging from 2–5 km/s. The number of Voronoi cells can vary between 4 and 500 cells, and the noise hyperparameters have a prior range from 0.001–0.5 s. Using 200 Markov chains in parallel, we sampled 310,000 models each until the convergence of the chains (Supplementary Figure

3). We thinned the chains by only retaining every 200th model in the chain to reduce the number of correlated samples. Each Markov chain starts with a different, random model drawn from the prior distribution.

Fig. 7 displays mapview slices through our 3D seismic velocity model at different depths and its standard deviation.

A major feature is the high shear wave velocities in the northeastern part of the model, which spatially coincides with the limestone formations (turquoise area in **Fig. 1**). The adjacent geologically complex areas of calcareous red greywacke and the alterations are marked by lower seismic shear velocity. In the middle part of the model, the seismic velocity increases with depth but shows horizontal variations as well.

In the horizontal cross-sections (**Fig. 8**), a trough is visible in the east part of the model (Lines 5 and 6) with fast velocities on the edges. On the contrary, in the western part of the model (lines 1–3), the trough is less pronounced.

In **Figs. 9**, we compare our 3D average shear wave velocity model from our passive seismics with active reflection seismics, conducted along the red lines in **Fig. 1** by Teck. There is an overall good agreement between the position of reflectors seen in the active reflection seismics

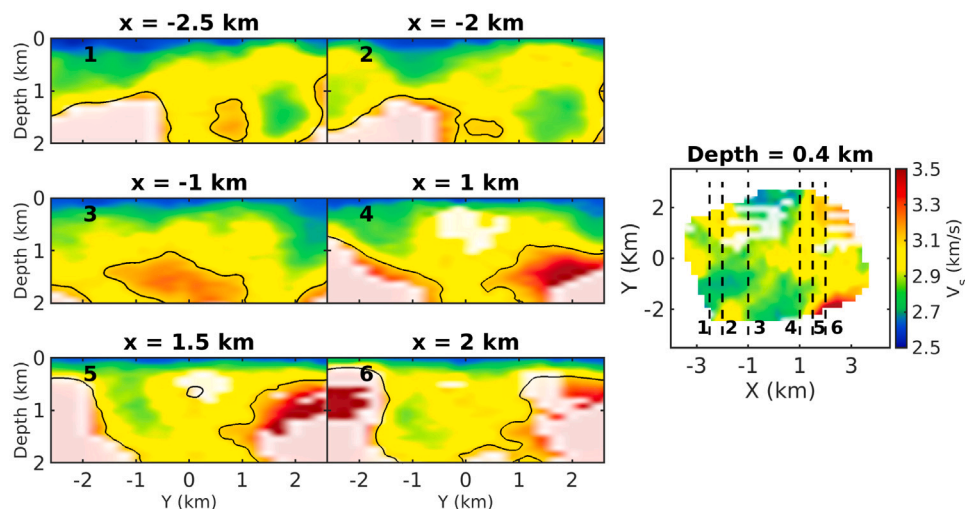


Fig. 8. Slices through the average 3D shear wave velocity model along the y -axis. Areas outside the station network with a standard deviation larger than 0.65 km/s are less color saturated. The black solid line marks the contour of $V_s = 3.1$ km/s.

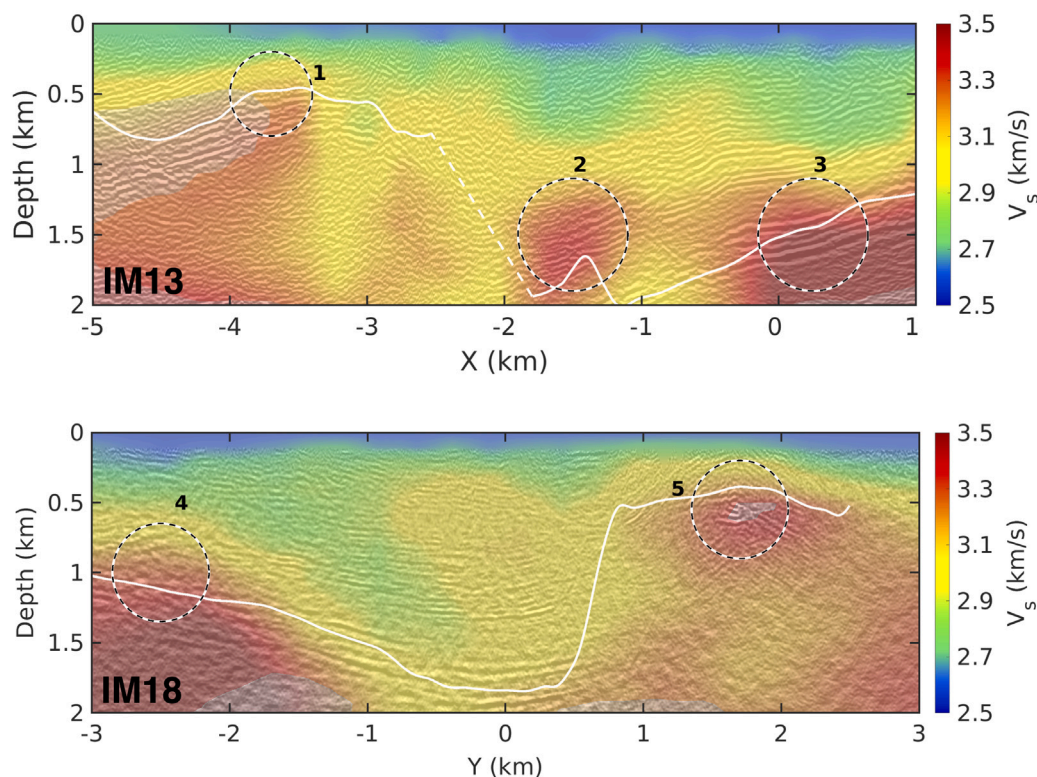


Fig. 9. Overlay of reflection seismics (gray shaded) with slices through the 3D seismic model along the profile lines IM13 and IM18 (Fig. 1). Circles highlight areas discussed in the text. The white line outlines the deep Pale Beds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and changes in seismic velocity, marked by circles. The Pale Beds (white line) seen in the reflection seismics are also clearly visible in the passive seismic model. The two reflectors (4 and 5) in the second profile are also clearly visible in the passive seismic imaging as high shear wave velocities.

4. Discussion and interpretation

At Navan, over 97% of the ore is hosted by a Courcayan, shallow-water succession termed the Pale Beds (Anderson et al., 1998; Redmond, 2023). The mineralization in the Ballinalack deposit occurs

in beds stratigraphically equivalent to those that host the Navan deposit (Jones and Brand, 1986). Therefore, Pale Beds are most likely to host base metal in the Collinstown study area. From previous work in our study area, the Pale Beds horizons have been interpreted from borehole data as reflectors in 2D reflection seismics. Fig. 10 displays a graphic log of a drill core from a borehole in the study area. The Pale Beds lie at around 800 m depth.

Hence, in terms of the usefulness of our passive method for mineral exploration, we want to test if (1) our model picks up the Pale Beds structure and (2) its depth agrees with those in the preexisting models for Collinstown from active methods.

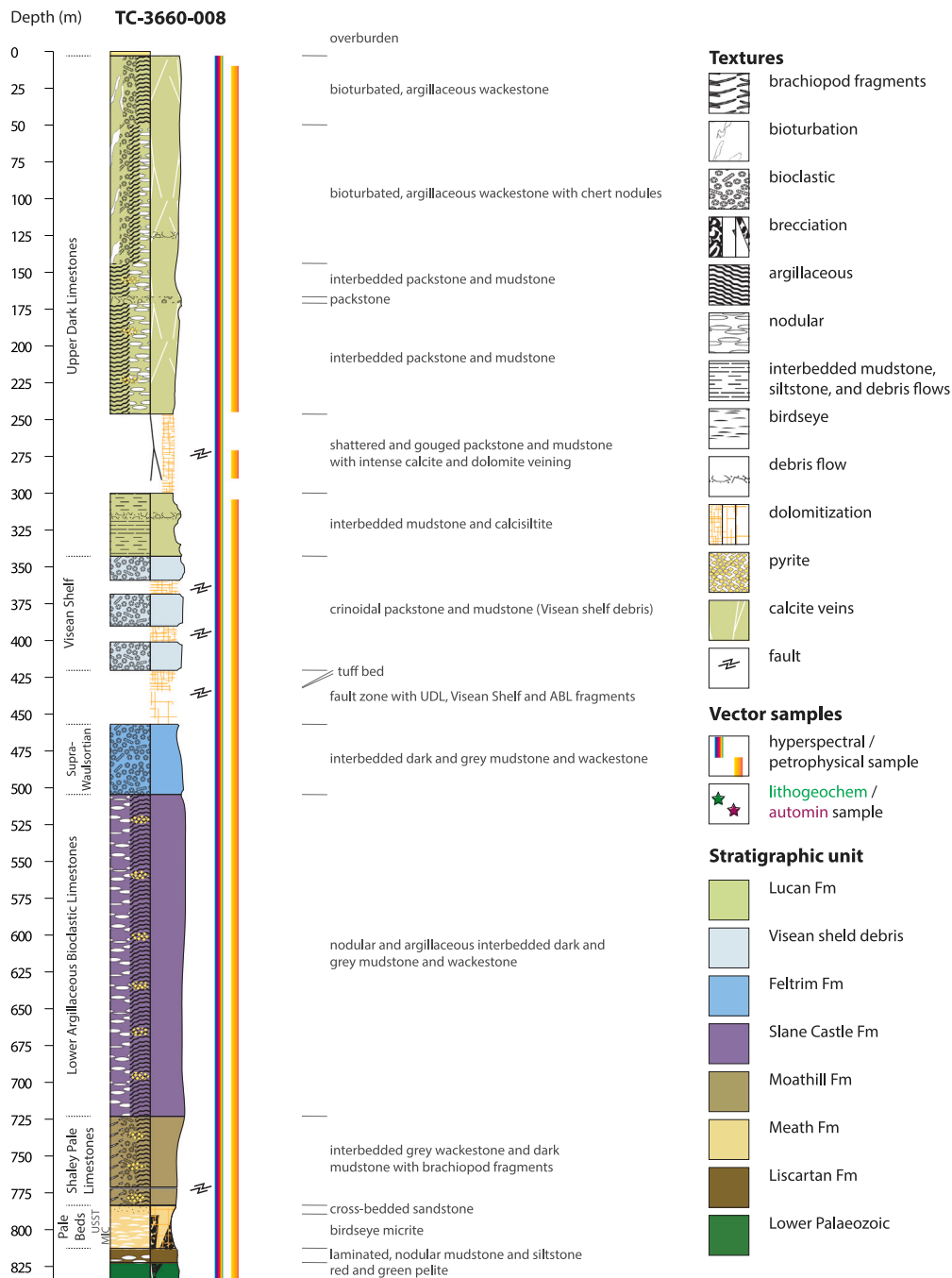


Fig. 10. Graphic log of a drill core at borehole TC-3660-008 (Fig. 1), logged within the VECTOR project.

Fig. 11 displays the main geological structures in Collinstown together with the reflection seismics, and an isosurface of $V_s = 3.1$ km/s derived from our 3D model. The argillaceous bioclastic limestone (ABL) overlays the Shaley Pales, covering the Pale Beds. This isosurface visually seems to coincide with the Pale Beds.

For a better comparison between the two surfaces, we calculate the depth from the surface to the isosurface in our 3D model and the depth of the Pale Beds along the two lines of the active reflection seismic profiles (Fig. 12, top). In this comparison, we used the deeper layer of the Pale Beds, as it better coincides visually with our isosurface. Along the line IM18, the depth of the isosurface agrees well with the depth of this particular geological structure, except for the deepest part, which our model underestimates a bit (red dots). Along the line IM13, our model picks up well the shallow and the deepest part of

this structural layer. The missing points in the middle of the line are because the terrace structure of the Pale Beds does not cover this area. The difference between the depth of the Pale Beds and our isosurface (Fig. 12, bottom) displays that we are slightly overestimating the depth by around 300 m along the right-hand side of the model, and underestimating it on the left-hand side by around 400 m. At the edges, however, where the raycoverage is less dense (Fig. 5), the resolution is worse, and vertical smearing results in larger differences between the Pale beds and the isosurface. Along the profile IM18 of the reflection seismics, our model matches with the Pale Beds ± 400 m, and at two points the difference is only 2 m and 72 m, respectively (arrows in Fig. 12, bottom). This comparison demonstrates that our model, though not as high resolution as active reflection seismics, picks up the Pale Beds at the right depths. Contrary to the 2D reflection seismics, our model

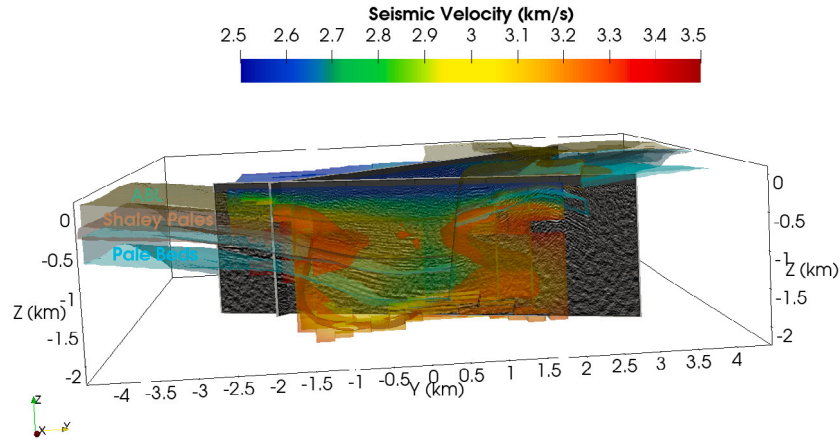


Fig. 11. Isosurface through the seismic shear wave velocity model (red) with $V_s = 3.1$ km/s, with geological layers and reflection seismic profiles. ABL: argillaceous bioclastic limestone. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

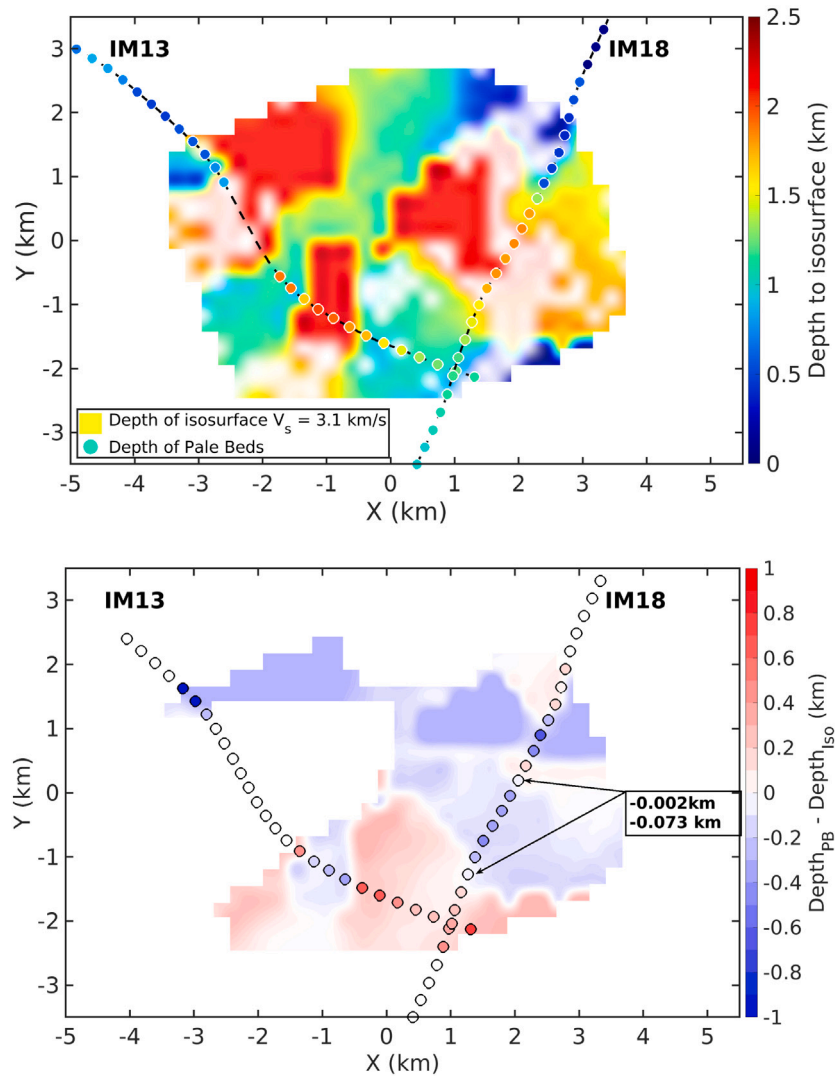


Fig. 12. Depth of the isosurface $V_s = 3.1$ km/s from the surface in our model, overlain by the depth of the Pale Beds along the two lines of the active reflection seismic profiles. Areas with a standard deviation above 0.6 km/s are more transparent (top). Difference between the depth of the Pale Beds and the isosurface (velocity) in the whole study area and along the two profile lines (bottom). The gap on the left results from the missing Pale Beds and our model space limitations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is 3D and hence allows us to image the Pale Beds location between the existing reflection seismic profiles. From this point of view, a low-cost passive seismic survey can be a good starting point before starting a 2D or even 3D active, potentially expensive active reflection seismic survey.

5. Conclusions

We have tested the usefulness of passive seismic methods in an exploration context in the Irish Midlands, which is known for its significant zinc deposits. Our seismic network, consisting of 206 vertical component geophones and nine broadband sensors, recorded ambient noise data for six weeks. We extracted fundamental Rayleigh wave group velocities from the data and used a probabilistic approach (Zhang et al., 2018, 2020) to invert for 3D subsurface seismic velocity. Our 3D model extends to a depth of 1.6 km, with a grid spacing of 125 m in the horizontal directions and 100 m in depth.

We compared our 3D passive seismic model with the results from two 2D reflection seismic profiles from a previous survey conducted in the area by Teck. Our passive method directly images shear wave velocities, while the active method highlights P-wave velocity contrasts. Nevertheless, we found that our passive method can pick up the main structures at depths and that the depth of geological layers, which are the most prospective host rocks for base metal mineralization, agrees well with those derived from the active 2D methods. Certainly, the passive method has lower resolution than the active reflection seismics. However, our method is passive (it only uses seismic signals that are present in the survey area), low-cost, and environmentally friendly. Additionally, we have demonstrated that extending from the 2D model to a 3D model with comparatively low cost and effort is possible. These encouraging results may open new directions in exploration strategies where 3D passive seismic methods are used first to explore the subsurface layering at a lower resolution and complement the following with higher resolution, expensive 2 or 3D reflection seismics surveys.

CRedit authorship contribution statement

Corinna Roy: Writing – original draft, Visualization, Validation, Formal analysis. **Trond Ryberg:** Validation, Supervision, Software, Project administration, Funding acquisition, Conceptualization. **Christian Haberland:** Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Kacie Wellington:** Validation, Resources. **Conor Moynihan:** Validation, Resources. **Philip Rieger:** Validation, Resources. **Murray W. Hitzman:** Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Corinna Roy reports financial support was provided by EU. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jappgeo.2025.105991>.

Data availability

The seismic data is available in the GEOFON database ([doi:10.14470/2J882031](https://doi.org/10.14470/2J882031)).

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